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VI SEMESTER

Experiential Learning - EVEN 2025-26

SDG 9: Industry, Innovation and Infrastructure

Project Title: **SCOUT** : Supply Chain Outage Understanding & Tracker

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Contents

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Introduction

"Global disruptions are rising across supply chains"

SUPPLY CHAINS DON'T FAIL SUDDENLY,THEY FAIL SILENTLY

Global Scenario

- **Rising geopolitical tensions & climate risks**
- **Price spikes, shipping delays, shortages**

The gap

- **No real-time visibility for SMEs**
- **Decisions made after disruption**

Our Approach

- **Continuous global monitoring**
- **Personalized early alerts**
- **Proactive decision-making**

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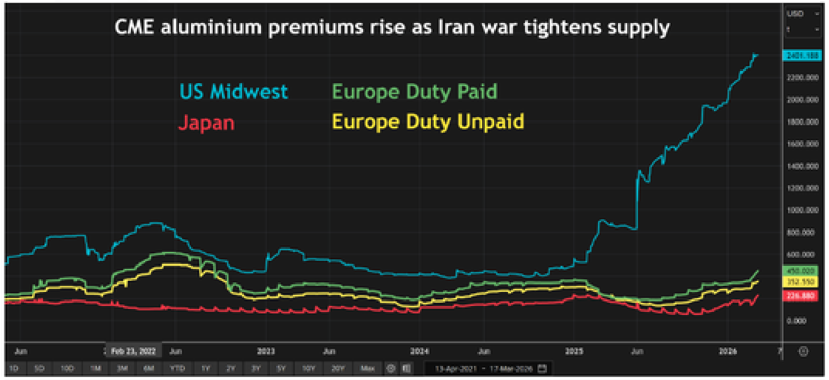
Supply Chain Disruption May Reduce Fertiliser Output By 15%, Raise Govt's Subsidy Bill By Rs 25,000 Cr

Energy disruption due to the West Asian war could reduce India's fertiliser production by 10–15 percent and raise the government's subsidy bill by Rs 25,000 crore, according to Crisil. Manufacturers may also face reduced profitability

Commentary Commodities

Iran war rattles the global aluminium supply chain



CME global aluminium premiums

From beer to cosmetics, Asia feels full force of war-fuelled energy crisis

By **Hyunjoo Jin**, **Daewoung Kim** and **Sophie Yu**

March 26, 2026 2:46 PM GMT+5:30 · Updated March 27, 2026

Summary Companies

- Asia most vulnerable to Middle East supply disruptions
- South Korean businessman says suppliers raising prices up to 50%
- Cosmetics container maker scrambling to secure plastic resin
- Consumers rush to stock up on rubbish bags, noodles

Tech Industry

AI memory crunch forces DRAM market into 'hourly pricing' model, report claims — small and medium-sized businesses fighting for survival

News By Luke James published March 3, 2026

Over 190,000 small and mid-size electronics companies are being squeezed out of the memory market by AI.

Why are memory cards in crisis?

Back in 1998, Sony introduced the Memory Stick. About 10 years later, it jumped on the SD and SDHC bandwagon, bringing these formats to the masses alongside its own media (Image credit: Sony)

AI is 'reshaping' many areas of photography – and memory cards are now feeling the impact.

ETPrime

Global memory-chip shortage sends prices soaring, strains tech industry

By Anupam Nagar, ETMarkets.com • Last Updated: Dec 03, 2025, 12:18:00 PM IST

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Synopsis
A global memory-chip shortage is disrupting the tech sector as AI firms, smartphone makers, and electronics companies battle for limited supplies. Prices for DRAM, flash, and HBM have surged, squeezing inventories and pushing up consumer device costs. Experts warn the crunch may persist until 2027, delaying major digital infrastructure projects.

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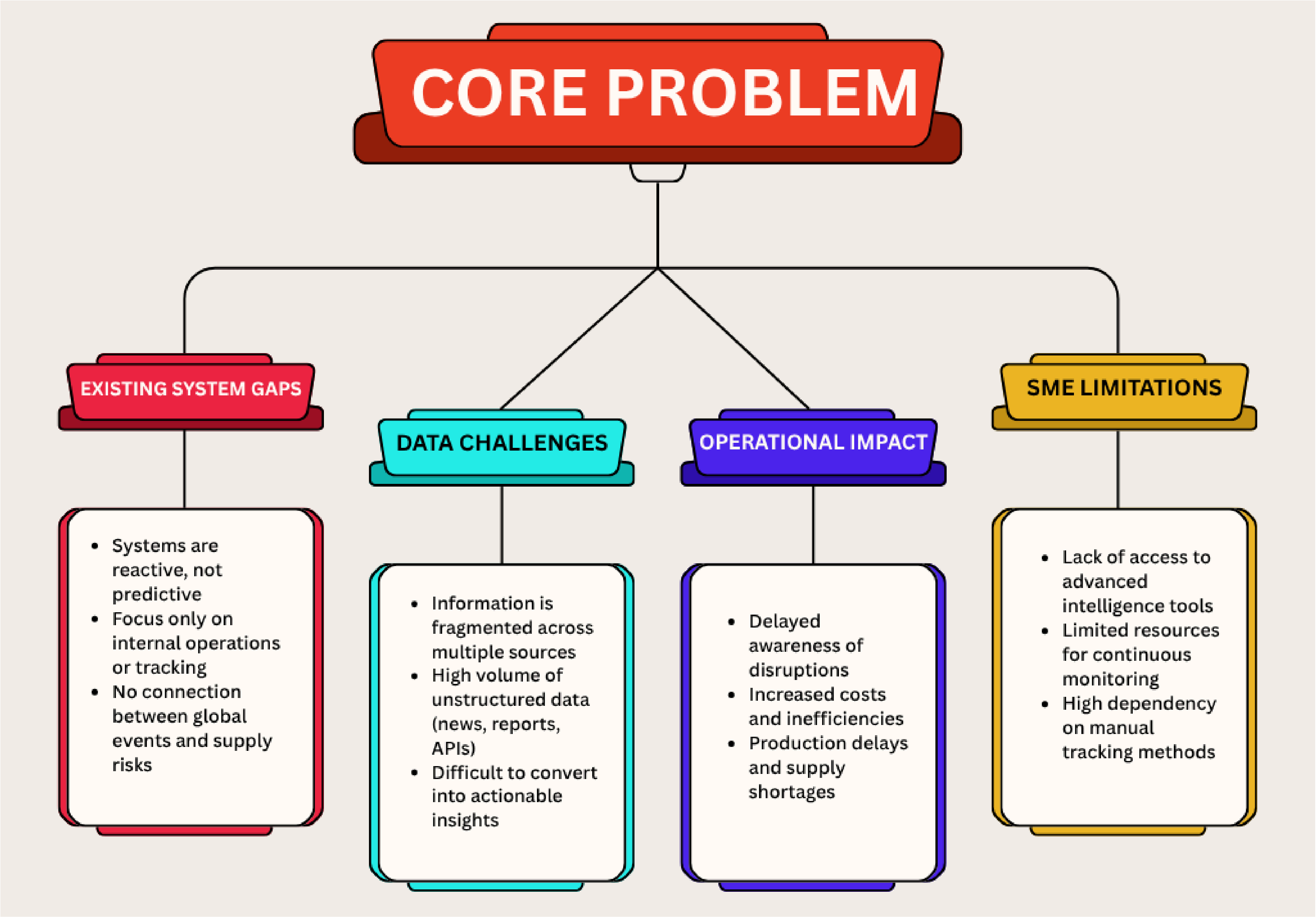
Existing System

	EXAMPLES	PROS	CONS
ERP Systems	SAP, Oracle SCM	• Centralized operations • Efficient resource planning	• No external risk monitoring • Reactive, not predictive
Visibility Platforms	Project44, FourKites	• Real-time shipment tracking	• Only tracks movement • Cannot predict disruptions
Market Intelligence Tools	Bloomberg, Refinitiv	• Rich global data	• Too much data (noise) • No personalization
Manual Monitoring (SMEs)	News, Excel	• Low cost	• Time-consuming • Misses critical signals

"EXISTING SYSTEMS INFORM BUT DON'T ANTICIPATE"

- React to disruptions instead of predicting them
- Provide data, but not decision-ready insights
- Built for enterprises, not adaptable for SMEs
- Fail to connect global events with specific supply chain risks

Identification of problem



Literature Review

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Paper title	Authors	Novelty features	Research gaps
Supply chain responses to global disruptions and its ripple effects: an institutional complexity perspective (2023) (Springer)	David M. Herold, Łukasz Marzantowicz (Springer)	Gives a strong theoretical lens for disruption response by explaining ripple effects through institutional complexity and competing logics across supply-chain actors. (Springer)	Mainly conceptual; it explains how disruptions are interpreted, but it does not provide a deployable detection pipeline, risk score, or real-time alert system. (Springer)
Detecting fake news and disinformation using artificial intelligence and machine learning to avoid supply chain disruptions (2023) (Springer)	Pervaiz Akhtar, Arsalan Mujahid Ghouri, Haseeb Ur Rehman Khan, Mirza Amin ul Haq, Usama Awan, Nadia Zahoor, Zaheer Khan, Aniqa Ashraf (Springer)	Shows how AI/ML can filter misinformation from multiple data sources so that false narratives do not trigger bad supply-chain decisions. (Springer)	Focuses on misinformation detection, but does not fully solve multi-source event fusion, supplier-specific personalization, or downstream disruption scoring. (Springer)
Event Identification for Supply Chain Risk Management Through News Analysis by Using Large Language Models (2024) (Springer)	Maryam Shahsavari, Omar Khadeer Hussain, Morteza Saberi, Pankaj Sharma (Springer)	Introduces LUEI/CERIA, a lightweight approach that learns related phrases from an event name and uses them to identify contributing events in news, which is very close to your early-warning use case. (Springer)	Still centered on news-based event identification; broader supplier-network mapping, quantitative risk ranking, and action recommendations are not the main focus. (Springer)
Using Sentiment Analysis to Detect Disruptive Events in Supply Chains (2024) (ScienceDirect)	Kiran Katoor Vishnuthilak, Benjamin Rolf, Tobias Reggelin, Sebastian Lang (ScienceDirect)	Proposes a framework that classifies events by risk type, geography, frequency, and sentiment, and uses transfer learning on news headlines to detect disruptive relevance. (ScienceDirect)	Works mainly at the headline/sentiment level; it does not deeply model supplier dependency, multi-tier propagation, or structured event knowledge. (ScienceDirect)
A sustainable inventory management model for closed loop supply chain involving waste reduction and treatment (2025) (Directory of Open Access Journals)	Giuseppe Aiello, Cinzia Muriana, Salvatore Quaranta, Islam Asem Salah Abusohyon (Directory of Open Access Journals)	Extends the **newsvendor / inventory** model into a **closed-loop, circular-economy** setting using the 4R concept and waste-treatment logic. (ScienceDirect)	Very useful for sustainability, but it is about **inventory and waste optimization**, not real-time disruption monitoring or supplier-risk prediction. (ScienceDirect)

Literature Review

Paper title	Authors	Novelty features	Research gaps
Artificial Intelligence Opportunities for Resilient Supply Chains (2024) (ScienceDirect)	Funlade Sunmola, George Baryannis (ScienceDirect)	Provides a strong **AI-for-resilience roadmap** and frames resilience using a **4-C model**: context, capabilities, choices, contingencies. (ScienceDirect)	It is a **roadmap/review**, so it does not itself implement a working pipeline for disruption detection, scoring, or alerting. (ScienceDirect)
Post-hazard supply chain disruption: Predicting firm-level sales using graph neural network (2024) (Science Explorer)	Shaofeng Yang, Yoshiki Ogawa, Koji Ikeuchi, Ryosuke Shibasaki, Yuuki Okuma (Science Explorer)	Uses **graph neural networks** plus internal/external factors and inter-firm relations to predict **post-hazard sales change**, with explainability analysis. (Science Explorer)	It is **post-disruption impact prediction**, not early warning; it is more useful for damage estimation than pre-event monitoring. (Science Explorer)
A Market Convergence Prediction Framework Based on a Supply Chain Knowledge Graph (2024) (MDPI)	Shaojun Zhou, Yufei Liu, Yuhan Liu (MDPI)	Demonstrates how a **supply-chain knowledge graph** can be used for prediction, which supports your idea of connecting suppliers, regions, commodities, and events in one semantic layer. (MDPI)	The paper is about **market convergence**, so it is not directly a disruption-alert system; it still needs event ingestion and risk prioritization to become operationally useful for your use case. (MDPI)
A system dynamics approach for leveraging blockchain technology to enhance demand forecasting in supply chain management (2025) (Directory of Open Access Journals)	SeyyedHossein Barati (Directory of Open Access Journals)	Uses **system dynamics** to model how blockchain adoption improves **data accuracy, transparency, and demand-forecasting performance** in supply chains. (Directory of Open Access Journals)	It is centered on **forecasting and blockchain transparency**, not on external disruption sensing, event classification, or multi-source risk intelligence. (Directory of Open Access Journals)
MCDFN: supply chain demand forecasting via an explainable multi-channel data fusion network model (2025) (Springer)	Md Abrar Jahin, Asef Shahriar, Md Al Amin (Springer)	Introduces **MCDFN**, a multi-channel CNN-LSTM-GRU fusion model with **explainable AI** tools such as ShapTime and permutation importance for demand forecasting. (Springer)	Strong for **forecasting**, but it does not solve the upstream problem of **detecting disruption events** from news or external signals. (Springer)

Problem Definition

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"Current SME supply chain systems lack an **integrated, real-time risk intelligence framework** capable of **ingesting heterogeneous global data** streams, performing contextual event detection, and generating predictive, supplier-specific disruption alerts. This results in delayed situational awareness, suboptimal decision-making, and increased vulnerability to dynamic geopolitical, economic, and logistical shocks."

Objectives

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- Construct a **Multi-Source Heterogeneous** Data Ingestion Layer with Schema-Agnostic Normalisation
- Build a Property Graph **Ontology Layer** to Model Supplier Exposure Through Multi-Hop Relationship Traversal
- Develop a Weighted **Composite Risk Scoring Engine** that identifies the Risk scores,Risk Paths and overall summary
- Deliver a **Mitigation Engine** that traces all the Risk Pipelines to suggest solutions.
- Deliver a **Manufacturer-Personalised Alert Interface** including Action recommendations and Alerts.

METHODOLOGY

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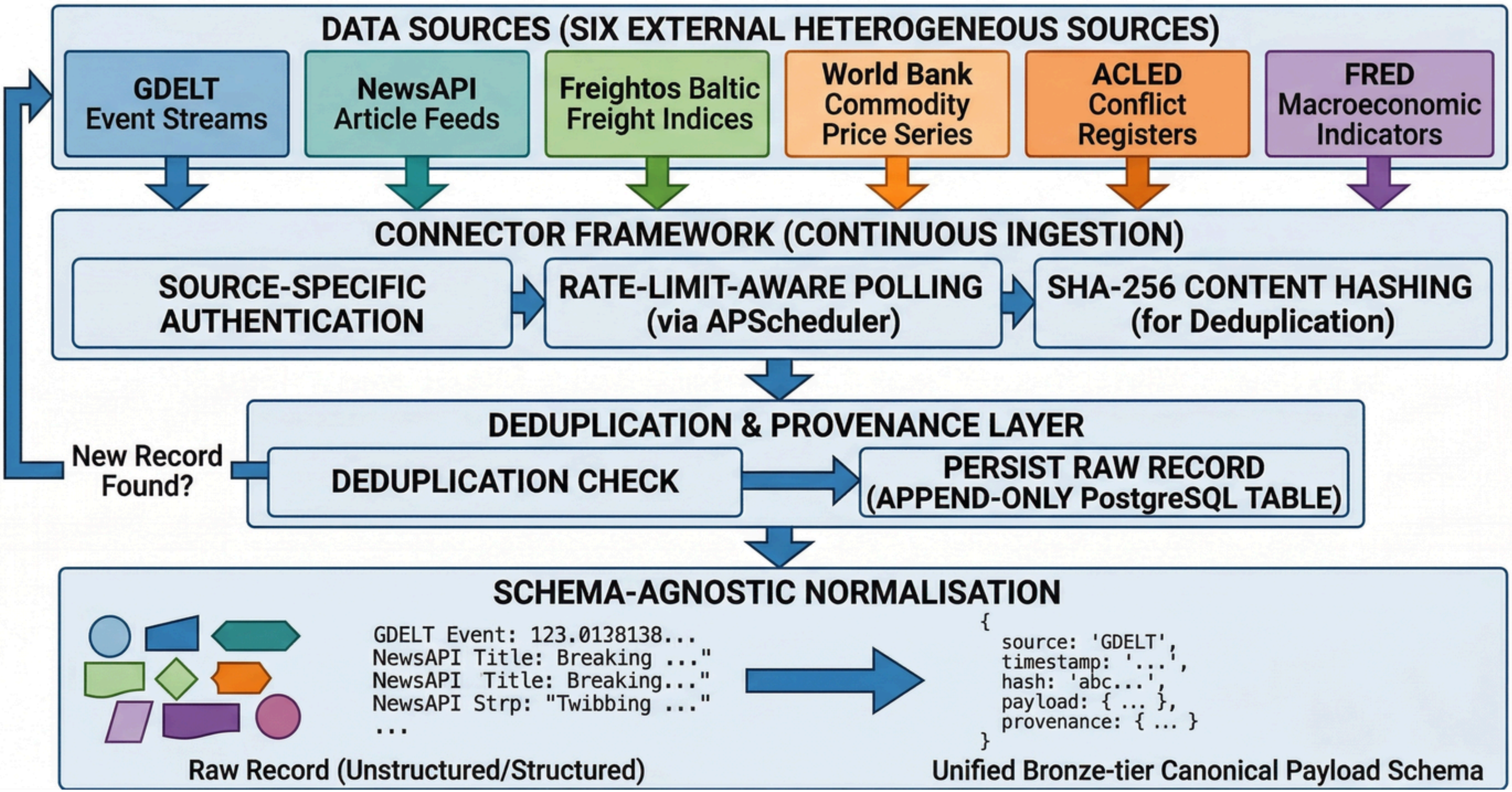
Methodology of 'Multi-Source Heterogeneous Data Ingestion Layer with Schema-Agnostic Normalisation'



METHODOLOGY

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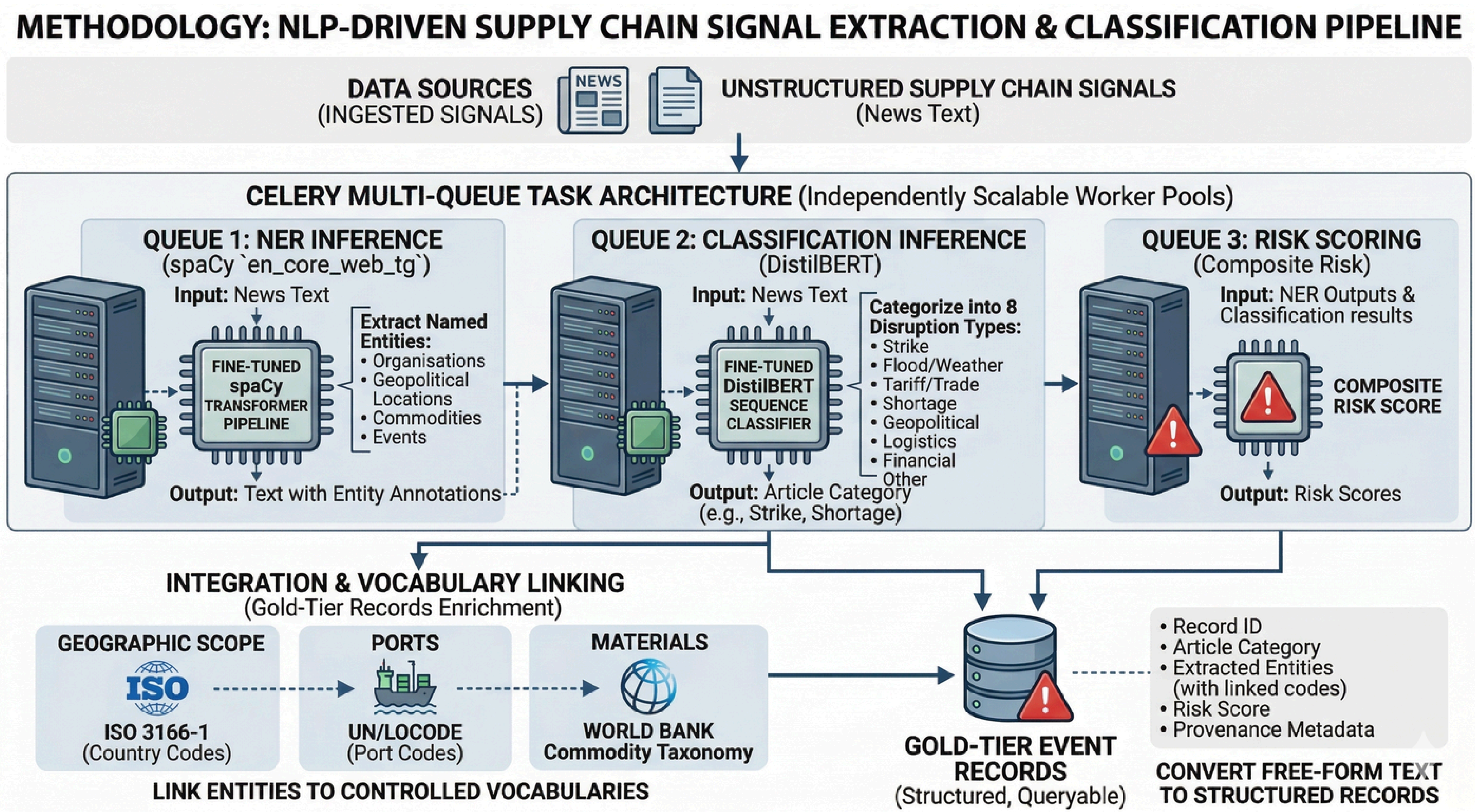

METHODOLOGY

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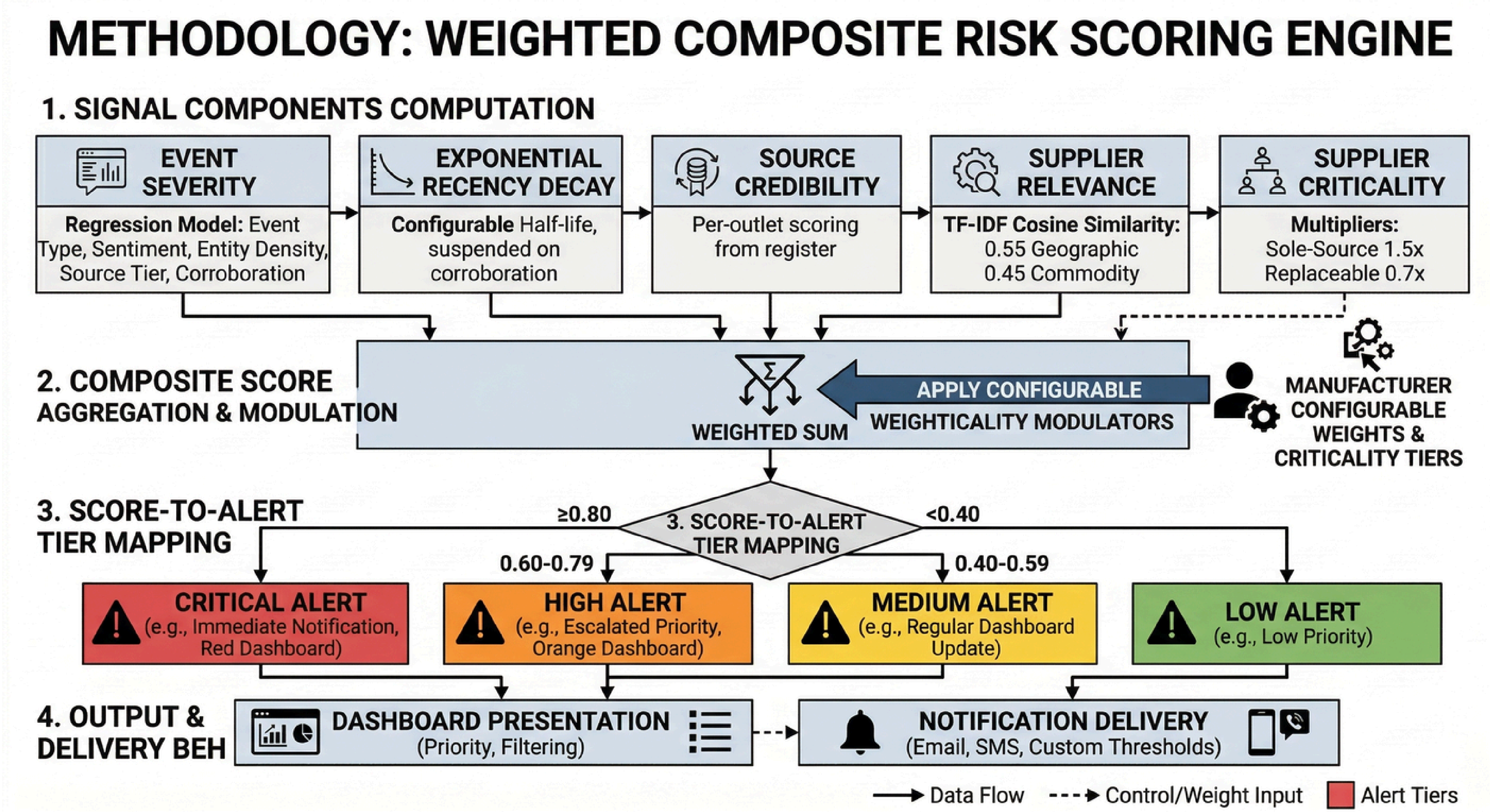

METHODOLOGY

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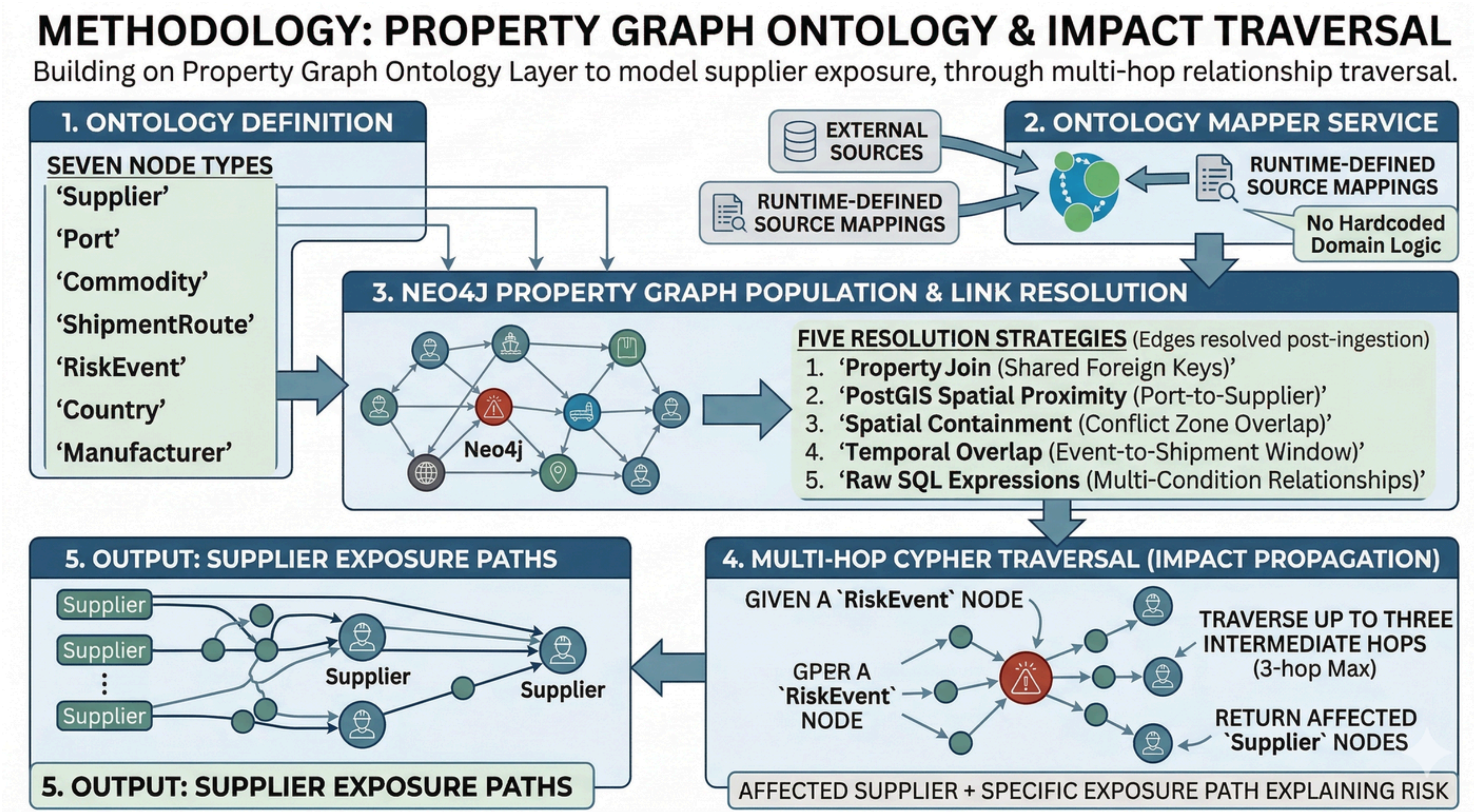

METHODOLOGY

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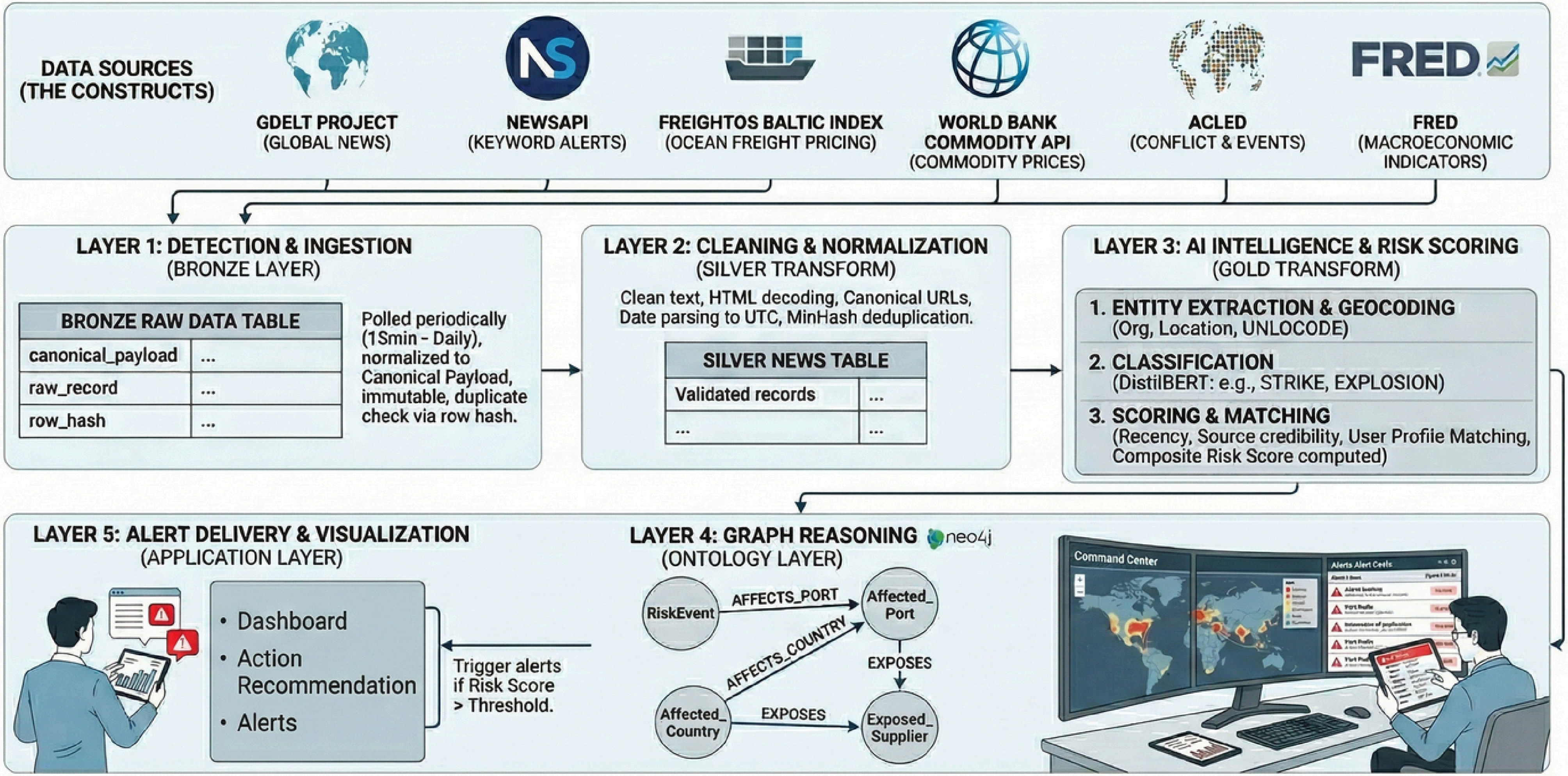

IMPLEMENTATION

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Data Source	Type of Data	What It Provides	Why It Is Useful in SCOUT	Update Frequency	Integration Method	Key Features Used
GDELT Project	Global events + news intelligence	Worldwide geopolitical events, protests, sanctions,	Core realtime event intelligence engine for disruption	Every 15 mins	DOC API + GDELT 2 Event/GKG feeds	EventCode, GoldsteinScale, AvgTone, locations,
ACLED	Conflict & instability data	Armed conflicts, riots, protests, political violence	Detects geopolitical instability affecting supplier regions	Daily/weekly	REST API / CSV	fatalities, conflict type, region severity
FRED (Federal Reserve Economic Data)	Economic indicators	Inflation, oil prices, trade indices, recession	Measures macroeconomic pressure impacting	Daily/Monthly	FRED API	CPI, oil prices, trade metrics, industrial indicators
Freightos Baltic Index	Shipping & freight analytics	Ocean freight rates, shipping cost trends, route	Detects logistics bottlenecks and transportation cost	Daily	API / scraped indicators	freight prices, route volatility
NewsAPI	Global news feeds	Business, economy, logistics, manufacturing	Secondary realtime news enrichment source	Near realtime	REST API	headlines, publishers, timestamps
SerpAPI Google News	Google News aggregation	Search-based news retrieval with broad coverage	Used for targeted searches like "semiconductor	Near realtime	SerpAPI	keyword-focused news retrieval
World Bank Open Data	Development & economic indicators	Trade flows, manufacturing output,	Useful for country-level risk weighting and long-term	Monthly/Yearly	World Bank API	GDP, logistics performance, trade dependency
Neo4j	Graph database	Relationship intelligence	Models supplier dependencies and multi-hop	Realtime	Neo4j Bolt driver	graph traversal, exposure paths
Databricks Community Edition	Big Data processing	Distributed ETL, historical analytics, scalable NLP	Bronze/Silver/Gold data lake processing	Batch/stream capable	Spark jobs	ETL, analytics, clustering
MinIO	Object storage	Raw file storage for CSV/ZIP/JSON archives	Stores raw ingestion data before	Persistent	S3-compatible API	Bronze data lake storage

Project Pipeline

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Global External Sources
↓
Real-Time Ingestion Layer
↓
Raw Event Storage (Bronze Layer)
↓
Normalization & Deduplication
↓
Structured Event Extraction (NLP Layer)
↓
Unified Event Storage (Silver Layer)
↓
Distributed Analytics & ML (Databricks)
→
Graph Intelligence Layer (Neo4j)
↓
Risk Propagation Engine
↓
Risk Scoring & Exposure Analysis
↓
LLM Reasoning & Mitigation Layer
↓
API Retrieval Layer
↓
Frontend Intelligence Dashboard
↓
Personalized Early Warning Alerts

Proposed tools and techniques

DATA INGESTION

- **Kafka**- real-time event streaming from all data sources
- **Spark-** batch processing and ETL pipelines
- **Flink-** stream processing for real-time data

DATA STORAGE

- **MinIO-** edge node local storage, works offline
- **Apache Parquet-** columnar format for querying
- **Postgres-** metadata and object relationships
- **Apache Iceberg-** structured data store

BACKEND

- **Python-** data transforms and core backend logic
- **HuggingFace Transformers-** event classification and NLP modelling
- **Pytorch-** Deep learning model training framework

Expected Outcomes

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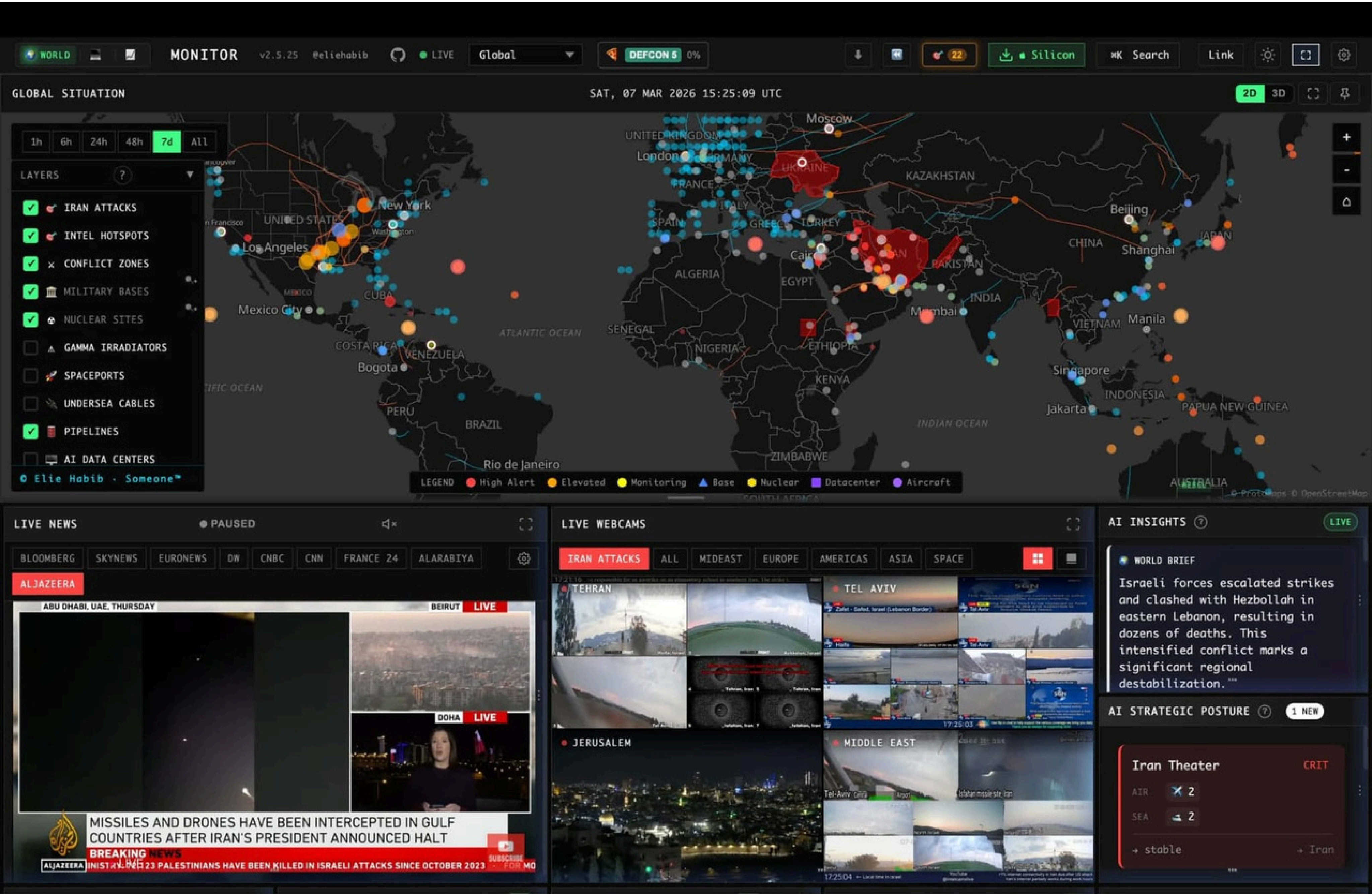
- **Automated Data Ingestion-** pulls from 6 sources such as GDELT, NewsAPI, Freightos Baltic Index, World Bank, ACLED, and FRED
- **NLP Based Event Detection-** extracts companies, countries, and commodities from raw news text
- **Supplier Profile Onboarding-** users register their key suppliers, raw materials, and sourcing regions
- **Personalized Risk Alerts -** cross-referenced against each user's registered suppliers, materials, and sourcing regions
- **Risk Scoring Engine**- ranks alerts by recency, source credibility, and geographic relevance
- **Severity Filtering-** lets users set alert thresholds to avoid notification overload
- **Weekly Email Digest-** summarized risk report delivered every Monday
- **Real-Time Risk Dashboard-** displays supply chain risk signals as a heatmap by supplier and geography
- **Fully Automated Pipeline-** ingest, clean, analyze, score, and alert which runs on schedule with no manual intervention


WORLD
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MILITARY BASES
NUCLEAR SITES
GAMMA IRRADIATORS
SPACEPORTS
UNDERSEA CABLES
PIPELINES
AI DATA CENTERS
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EURONEWS
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CNBC
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ALARABIYA
ALJAZEERA
ABU DHABI, UAE, THURSDAY
BEIRUT LIVE
DOHA LIVE
MISSILES AND DRONES HAVE BEEN INTERCEPTED IN GULF COUNTRIES AFTER IRAN'S PRESIDENT ANNOUNCED HALT
BREAKING NEWS
23 PALESTINIANS HAVE BEEN KILLED IN ISRAELI ATTACKS SINCE OCTOBER 2023
LIVE WEBCAMS
IRAN ATTACKS
ALL
MIDEAST
EUROPE
AMERICAS
ASIA
SPACE
TEHRAN
TEL AVIV
JERUSALEM
MIDDLE EAST
AI INSIGHTS
LIVE
WORLD BRIEF
Israeli forces escalated strikes and clashed with Hezbollah in eastern Lebanon, resulting in dozens of deaths. This intensified conflict marks a significant regional destabilization.
AI STRATEGIC POSTURE
1 NEW
Iran Theater
CRIT
AIR 2
SEA 2
→ stable
→ Iran

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Results Achieved

Throughput and Latency Performance:

HVE-OS achieves 5× higher throughput and 70% lower latency than traditional ETL pipelines, enabling faster real-time processing

Memory Pressure and Query Complexity:

The modular architecture significantly reduces memory overhead and scales efficiently with increasing graph complexity.

Concurrency and Thread Offloading:

Background thread-pool offloading minimizes main-thread blocking and maintains consistent system responsiveness under heavy workloads.

Risk Calculation & Routing:

The risk engine computes supplier-specific disruption scores and automatically categorizes entities for real-time risk monitoring and decision support.

Results Achieved

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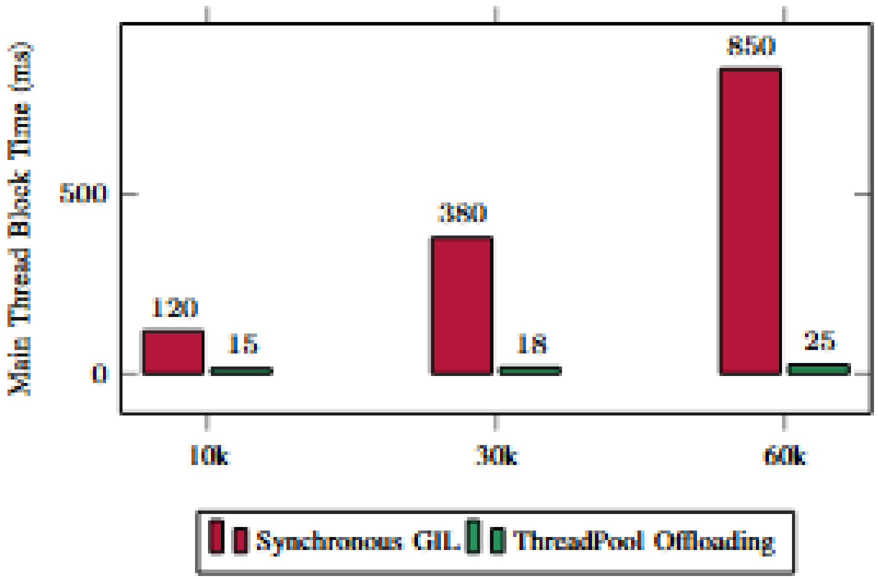

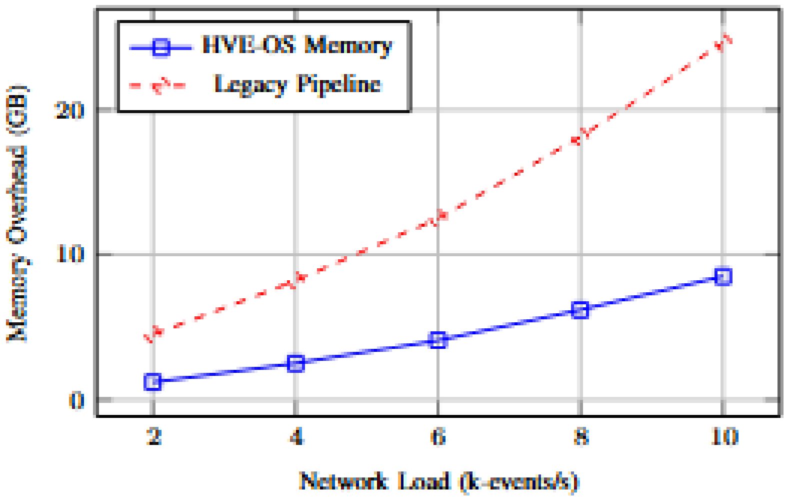

PERFORMANCE BENCHMARKS: HVE-OS VS. TRADITIONAL ETL

Metric	Static ETL Baseline	HVE-OS (DAG)
Max Throughput (events/s)	2.1k	**10.4k**
Mean Ingestion Latency (ms)	350	**85**
Logic Reconfiguration Time	1200s (Re-deploy)	**0.5s (Hot-swap)**
AI Linking Accuracy	68%	**91%**
Graph Traversal Speed (N=4)	3.2s	**0.15s**

Conclusion

In conclusion, this proposed **Supply Chain Disruption Early Warning System** redefines how SMEs approach risk by shifting from reactive responses to anticipatory, intelligence-driven decision-making. By transforming fragmented global signals into timely, actionable insights, it empowers businesses to detect disruptions before they escalate, minimize their impact, and maintain operational continuity. Ultimately, the system aims to build resilient, adaptive supply chains capable of thriving in an increasingly uncertain and dynamic global environment.

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Thank You